

Chenghao (Tommy) Jiang

3D Computer Vision · SLAM · Generative Models

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EDUCATION

University of Wisconsin-Madison

MS in Electrical and Computer Engineering

- GPA: 3.82/4.0

Sep 2024 - Dec 2025

Madison, WI

University of Manchester

MS in Communication and Signal Processing

- GPA: 83.5/100 · Distinction Honor

Sep 2022 - Dec 2023

Manchester, UK

Changchun University of Science and Technology

BEng in Optoelectronic Information Science and Engineering

- GPA: 3.86/5.00 · Rank: 10/221

Sep 2018 - Jun 2022

Changchun, China

EXPERIENCE

Tera AI

3D Vision Researcher

Remote

Feb 2026 - Present

- Built a geometry-guided data pipeline to derive high-confidence geometric pseudo-labels for dense image correspondence from internally collected flight video and prior scene geometry.
- Fine-tuned UFM on internal evaluation pairs, reducing average endpoint error by 67.5% for camera-to-map matching (12.40 to 4.03 px) and 31.7% for frame tracking (5.33 to 3.64 px).
- Developed flight-replay diagnostics linking correspondence failures to downstream pose processing and geographic priors; the UFM frontend raised matching throughput 5x (4 to 20 FPS) versus RoMa and helped reduce GPS-referenced trajectory RMSE from >10 m to approx. 3 m.
- Implemented and benchmarked 3D reconstruction pipelines spanning classical SfM, feed-forward 3D models, and Gaussian Splatting; evaluated Sim(3)-aligned point clouds by point-to-mesh distance and novel-view quality by held-out photometric error.

Research Intern - Part-time

Aug 2025 - Feb 2026

- Prototyped windowed deployment of SStream3R for long-horizon flight video; identified memory growth and clip-level latency as blockers for real-time, edge-constrained localization, motivating an online correspondence-based frontend.

Johns Hopkins University

Research Assistant · Advisor: Prof. Anand Bhattad

Sep 2025 - Feb 2026

Remote

- Developed a ViT-based latent encoder that disentangles illumination from scene content; on the MIT Multi-Illumination test set, reduced color-corrected relighting RMSE by 7.2% (0.222 to 0.206) and improved SSIM from 0.571 to 0.600 over the published latent-intrinsics baseline.
- Designed a DiT-based generative pipeline using illumination as a prompt and reference scene content as control for high-quality, controllable image relighting.

The Hong Kong University of Science and Technology (Guangzhou)

Research Assistant · Advisor: Prof. Haoang Li

Nov 2023 - Jun 2024

Remote

- Developed an animatable human Gaussian Splatting pipeline using a canonical avatar representation and SMPL-driven deformation across per-frame body poses.
- Co-designed and implemented a correspondence-guided feature-consistency loss using RoMA matches and DINO features across viewpoints and body poses, complementing photometric supervision to enforce part-level appearance consistency under articulated motion.

SELECTED PROJECTS

Privacy-Aware Sensor Data for Cooperative Perception

Supervised by Prof. Akarsh Prabhakara

Jun 2025 - Jul 2025

Madison, WI

- Explored cooperative SLAM under privacy constraints using SHARP, transmitting pointmap-based novel-view renderings instead of raw images.
- Evaluated VGGT on the OPV2V dataset across ego-only, SHARP-generated, and raw multi-agent inputs.
- Extended the CARLA simulation in OPV2V with depth sensing for point-cloud rescaling and downstream 3D recovery.

PUBLICATIONS

Privacy-Aware Sharing of Raw Spatial Sensor Data for Cooperative Perception

Bangya Liu, Chenghao Jiang, Chengpo Yan, Suman Banerjee, Akarsh Prabhakara

HotMobile 2026

Under review

SKILLS

Programming

Python · CUDA · PyTorch · LaTeX

Research

3D reconstruction · SLAM · Gaussian Splatting · Diffusion Models · VLMs